Department of Computer Science and Engineering
BRAC University

Set A

Examination: Midterm | Semester: Fall 2025
Duration: 1 hour 30 minutes | Full Marks: 30

CSE 423: Computer Graphics

Name:	ID:	Section:

Answer the following questions. Understanding the questions is part of the examination. The numbers on the right margin indicate marks.

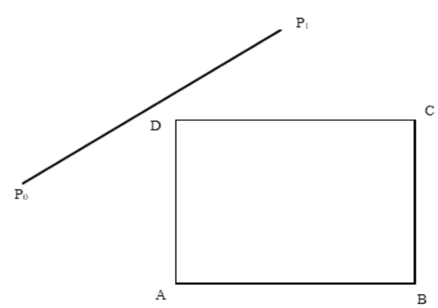
Question 1 [CO1]

a. A digital mapping system utilizes the **Digital Differential Analyzer (DDA)** algorithm to render objects by plotting **discrete** pixel points along a line. When the view is zoomed in, the objects appear **jagged** along their diagonal edges. Without modifying the hardware, **propose** any method that could be applied to minimize this jagged visual effect. Now, if the system used a **continuous beam drawing** method instead, would the DDA algorithm be necessary to generate these lines? In two or three sentences, **explain** why or why not. **2+2**

b. Suppose, for your CSE423 Lab Assignment 1, you need to draw a slanted line with starting point (220, 111) and ending point (111, -111). However, instead of using *GL_LINES*, you need to draw the line using the Midpoint Line (MPL) Drawing algorithm. Now, using the MPL Algorithm, **determine** the **first six** pixels (including the starting pixel) of this line. Show the necessary calculations in a table. **6**

Question 2 [CO2]

a. You are given a rectangular clipping window **ABCD** and a line segment $\mathbf{P_0P_1}$ as shown in the figure. The line segment $\mathbf{P_0P_1}$ is clearly **outside** the clipping window **ABCD**. However, the ***Cohen–Sutherland line clipping algorithm*** cannot **directly reject** this line. Instead, it **subdivides** the line before rejection. **4**



Do you agree with this statement? **Explain** and **justify** your answer in two or three sentences.

b. Suppose a rectangular clipping window is defined as (-120, -100) and (120, 100). A line segment with endpoints (15, 90) and (100, -130) needs to be drawn on this window. Using a **parametric** line clipping algorithm, **compute** the clipped segment of this line that can be visible in the window. Show the necessary calculation in a table with the updated values of $\mathbf{t_E}$ and $\mathbf{t_L}$. Also **include** a simple drawing of the scenario showing the intersection points. (The drawing does not need to be exact; only a rough estimate is enough.) **6**

Question 3 [CO3]

Kaguya and Shirogane are the two top elite students of Shuchiin Academy. While studying affine transformations in R3, Kaguya presented the following three equations for a combined transformation:

$$x' = 10 + 7z - ax$$
$$y' = -9y - 5 - 3z$$
$$z' = cz$$

Kaguya stated that this single composite operation could be decomposed into three standard transformation matrices. He also mentioned that the matrix, which has missing values **a** and **c**, represents a **uniform scaling.** One of these three transformations is not a **linear transformation**. He named it $\mathbf{M_2}$.

a. **Identify** the values of **a** and **c**. Then, **identify** the three standard transformations referenced in the passage and **show** their **4x4** homogeneous matrices. **1+3**

b. Now, Shirogane needs to calculate the final position of a point after applying the three operations in a specific sequence: **2+4**

- Previously mentioned **Uniform Scaling** is applied first.
- Then, the transformation $\mathbf{M_2}$ is applied
- The **remaining transformation** is applied last.

Find out the composite transformation matrix formed by these 3 operations in the specific order. Then, apply this composite transformation to the point **P (2, 5, 10)** with respect to another point **(-2, 4, -1)**. **Determine** the final coordinates of the transformed point, **P'**.

*************** **The End** ***************